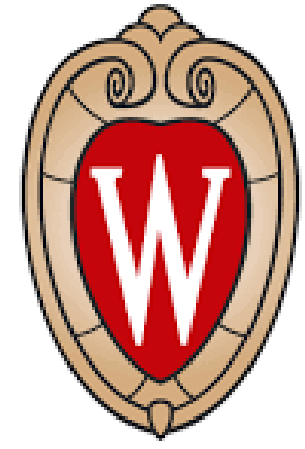


Effects of the Common Buckthorn on Opossum Activity across an Urbanization Gradient in Southern Wisconsin



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Introduction

Buckthorn changes the dynamics of an ecosystem, limiting the growth of native vegetation, changing soil composition, and changing habitat availability (Knight et al. 2007, Stewart et al. 2021). The fast-growing nature of buckthorn provides a dense habitat, protection, and less perceived predation risk for small mammals (Mattos and Orrock 2010, Guiden and Orrock 2017). We predicted opossums would be more likely to live and forage in these areas because they are more protected and have access to more food. Another factor that may influence opossum activity is urbanization. Due to anthropogenic food resources and possibly refuge from predators, opossums are more likely to be closer to people (Rodriguez et al. 2021). These animals are generalist feeders, meaning they can take advantage of both natural and anthropogenic food sources. We expect to see more opossum activity in urban environments.

Methods

To collect data, we set up motion-activated wildlife trail cameras at 16 sites. These sites were categorized as urban, rural, or agricultural environments. Each site had 2 plots, one containing invasive shrubs and one where invasive shrubs had been removed. We used Wildlife Insights, an AI platform used to analyze and scan images from trail cameras, to identify the animals in the photographs (Ahumada et al. 2020). We used program R to run a generalized linear mixed effects model to see if there was a significant difference between opossum activity between invasive and control plots. We also used the same test on the pooled data to see if there was a significant difference in the opossum activity in urban, rural, and agricultural environments.



Results

There was a trend of greater opossum activity on plots where buckthorn was removed ($\beta=1.05\pm0.65$ SE, p -value = 0.10), however, the difference was not significant. On average, there were 18 detections on plots with buckthorn removed and 7.6 detections on plots where buckthorn was intact.

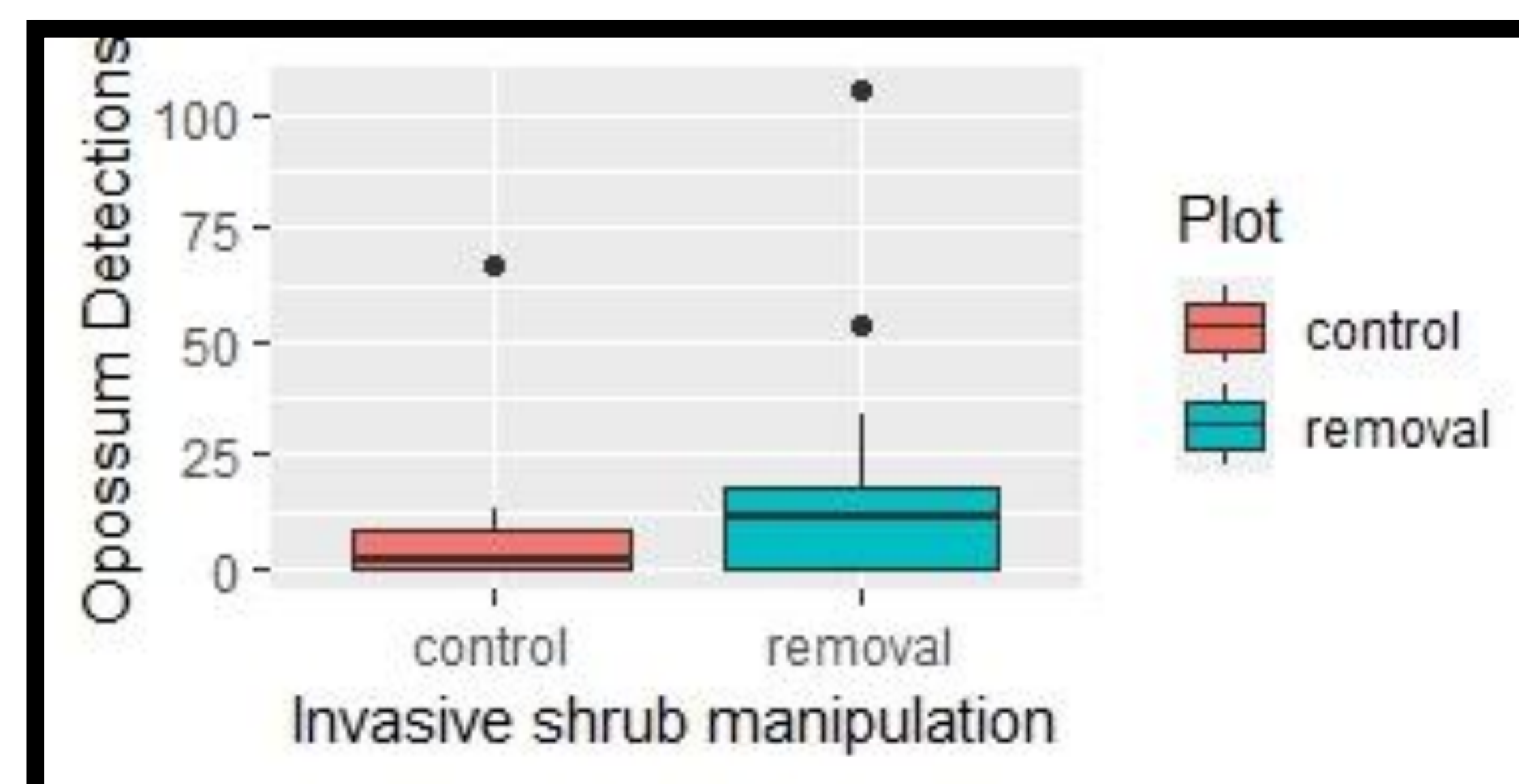


Figure 1. A comparison of the average number of detections of opossums in control plots and plots with buckthorn removed.

Although there was no significant overall effect of urbanization on opossum activity either ($df = 2$, p -value 0.092) there was a trend of higher activity in urban environments. There are 23.4 detections, on average, in urban environments, 5.5 in rural environments, and 8.8 in agricultural environments (Figure 2). There was no interaction between the habitat context (urban, rural, or agricultural) and buckthorn presence or absence (p -value = 0.666).

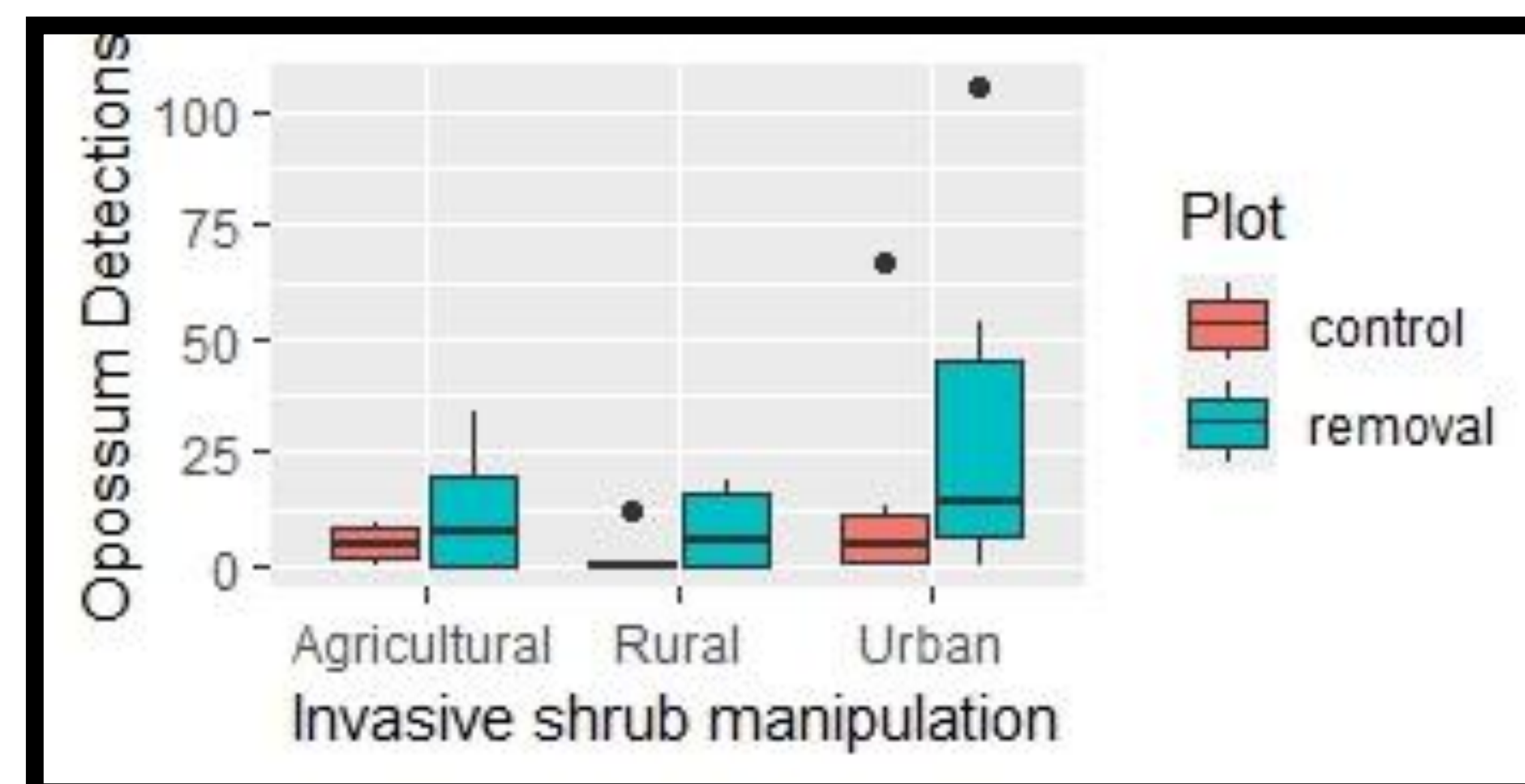


Figure 2. A comparison of the average number of detections of opossums in urban, agricultural, and rural plots.

Discussion/Conclusion

While we did observe strong trends in our data, some of the reasons we didn't measure significant differences could include:

- The detection method used, as the thickness of vegetation may have limited visibility
- How difficult foraging is in areas with different densities of vegetation for opossums, as it could have been harder due to the difficulty of hunting prey, or easier due to a higher density of prey
- Inconclusive assessments of how much anthropogenic food and water were available in the urban areas surveyed

The lack of significance of these differences could also tell us that buckthorn and urbanization do not have a big impact on opossum activity and that these factors have little to do with opossum conservation concerns. Further research on this topic is important to figure out if invasive species such as buckthorn impact habitat use, disease outbreaks in native biota, changes in biodiversity, and if more specific aspects of behavior such as foraging or mating are changed by these factors.

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